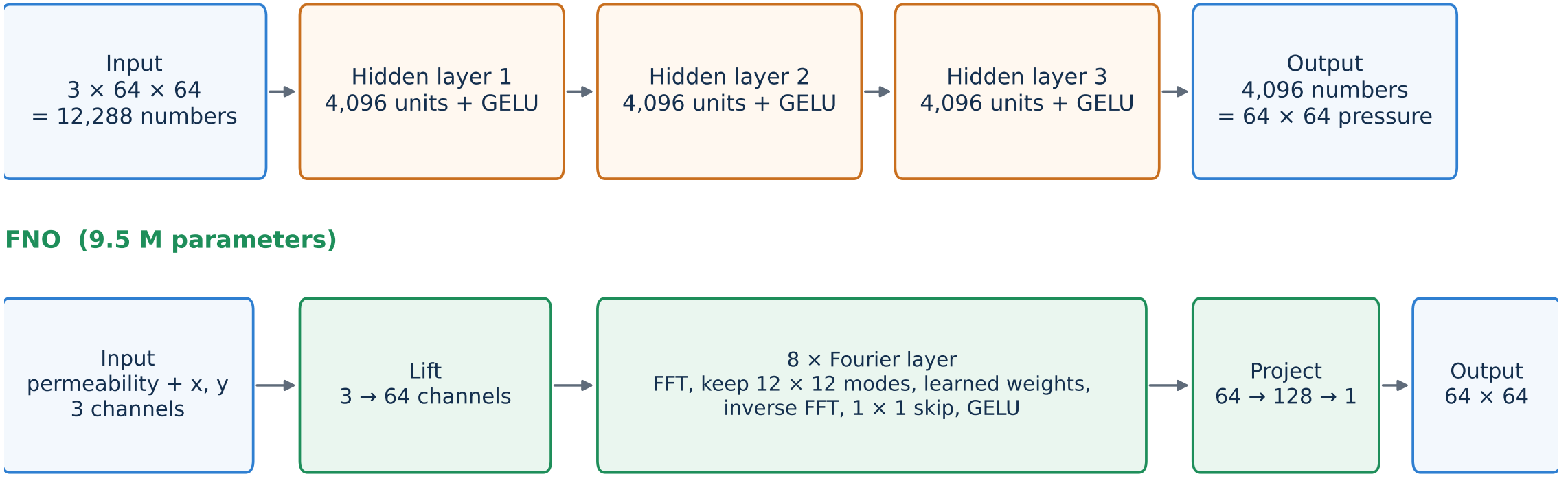
MLP baseline (100.7 M parameters)
Input
3 × 64 × 64
= 12,288 numbers
Hidden layer 1
4,096 units + GELU
Hidden layer 2
4,096 units + GELU
Hidden layer 3
4,096 units + GELU
Output
4,096 numbers
= 64 × 64 pressure
FNO (9.5 M parameters)
Input
permeability + x, y
3 channels
Lift
3 → 64 channels
8 × Fourier layer
FFT, keep 12 × 12 modes, learned weights,
inverse FFT, 1 × 1 skip, GELU
Project
64 → 128 → 1
Output
64 × 64